Apply the law of conservation of energy to analyse system efficiency in terms of energy inputs, outputs, transfers and transformations

Learning intention

- apply the law of conservation of energy to analyse system efficiency in terms of energy inputs, outputs, transfers and transformations.

Teach and model

- explaining efficiency and recognising that in energy transfer and transformation a variety of processes can occur.
- using and critiquing representations such as Sankey diagrams to show energy inputs, changes and outputs in a system.
- comparing the efficiency of electricity generation from coal and other sources such as nuclear, hydroelectricity, gas, solar and wind.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

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Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.